

CAO BANG, Viet Nam

WMO: 488080

Lat: 22.6636N

Lon: 106.2510E

Elev: 797

StdP: 14.28

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	43.2	46.0	35.7	31.6	53.7	38.9	35.8	52.5	15.3	67.9	12.8	66.1	1.6	180	0.282

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	95.1	80.9	93.3	80.3	91.6	79.8	82.8	91.7	81.9	90.4	81.0	89.0	5.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.5	163.1	88.7	79.6	158.4	87.1	78.9	154.6	85.8	47.7	92.2	46.7	90.8	45.7	89.4	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	10.8	8.9	39.1	99.4	3.0	1.7	36.9	100.6	35.1	101.6	33.4	102.6	31.2	103.8
			38.4	85.1	2.7	1.2	36.5	86.0	34.9	86.7	33.4	87.4	31.5	88.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.9	58.3	62.1	67.3	75.0	79.7	82.4	82.7	82.2	80.1	75.4	68.4	60.8
	DBStd	10.17	6.69	7.89	6.92	6.28	4.74	3.12	2.76	2.71	3.43	4.31	5.34	5.74
	HDD50	9	6	2	0	0	0	0	0	0	0	0	0	1
	HDD65	618	229	142	56	6	0	0	0	0	0	1	25	159
	CDD50	8377	263	340	537	751	921	973	1015	998	903	786	553	337
	CDD65	3511	21	60	128	307	456	523	550	533	453	322	128	30
	CDH74	30250	72	326	667	2286	4297	5441	5624	5021	3792	1993	615	116
	CDH80	11045	5	80	189	774	1677	2105	2165	1893	1408	607	130	12
(14) Wind	WSAvg	3.0	2.9	3.4	3.8	4.1	3.4	3.1	3.0	2.7	2.5	2.5	2.6	2.6
(15) Precipitation	PrecAvg	58.8	1.4	1.1	2.1	3.2	7.8	11.0	10.7	10.0	5.1	3.2	1.8	1.3
	PrecMax	70.2	4.9	3.1	8.5	7.2	12.4	18.1	21.4	14.1	10.2	9.6	5.4	4.1
	PrecMin	43.7	0.2	0.1	0.2	0.7	4.1	4.7	4.9	2.0	2.1	0.7	0.1	0.3
	PrecStd	7.4	1.1	0.9	2.1	1.6	2.2	3.7	3.1	3.2	2.4	1.9	1.4	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.5	87.0	90.3	96.4	97.8	96.8	96.6	95.5	93.9	90.9	86.8	81.6
		MCWB	68.3	70.7	73.5	76.4	80.5	81.6	82.1	81.2	79.5	76.5	74.5	70.5
	2%	DB	75.6	82.4	85.0	91.0	93.8	93.8	93.9	93.4	91.9	88.2	83.6	77.2
		MCWB	66.3	70.4	73.6	77.4	80.4	81.4	81.3	80.6	78.8	75.2	72.5	67.7
	5%	DB	72.1	78.1	81.1	87.6	91.0	91.7	91.9	91.4	90.0	86.0	80.7	73.5
		MCWB	64.2	68.6	71.9	76.2	79.0	80.7	80.8	80.2	78.1	74.1	70.3	64.9
	10%	DB	68.6	73.7	77.2	84.5	88.3	89.5	89.7	89.3	87.8	83.6	77.3	70.0
		MCWB	62.9	66.7	70.1	75.1	77.6	80.1	80.3	79.8	76.9	73.7	68.9	62.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.5	72.7	75.9	79.9	83.2	83.8	83.9	83.5	81.8	79.2	76.4	72.1
		MCDB	77.4	82.4	85.9	90.0	93.5	92.9	93.0	92.2	90.1	86.9	83.9	79.5
	2%	WB	67.3	70.9	74.2	78.2	81.4	82.6	82.6	82.2	80.6	77.3	74.6	69.2
		MCDB	73.9	79.9	82.6	87.7	91.2	91.3	91.4	90.8	88.7	84.7	80.9	75.0
	5%	WB	65.7	69.3	72.7	77.0	80.1	81.6	81.6	81.0	79.5	76.2	72.9	66.9
		MCDB	71.1	76.7	79.6	85.6	88.7	89.6	89.7	89.0	87.0	82.8	77.9	72.0
	10%	WB	63.8	67.6	71.1	75.8	78.9	80.6	80.7	80.1	78.5	75.1	71.1	64.3
		MCDB	67.8	73.2	76.5	83.1	86.1	88.0	88.0	87.2	85.3	80.9	75.2	68.5
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	12.4	11.8	12.9	13.9	13.0	12.9	13.6	14.8	14.9	15.4	14.4
		MCDBR	18.6	20.3	18.2	18.2	18.3	16.6	17.0	17.4	18.0	18.5	19.7	19.3
		MCWBR	9.0	8.9	7.7	6.9	6.7	6.0	6.0	6.2	5.8	6.7	9.0	9.7
	5% WB	MCDBR	15.6	18.2	16.4	16.1	15.9	14.7	15.1	15.8	15.4	15.2	15.4	16.3
		MCWBR	8.1	8.6	7.6	6.8	6.6	6.3	6.3	6.5	5.9	6.4	8.1	9.0
(40) Clear-Sky Solar Irradiance	taub		0.579	0.601	0.772	0.730	0.550	0.535	0.509	0.569	0.601	0.686	0.580	0.606
	taud		1.744	1.737	1.451	1.573	2.029	2.090	2.152	1.963	1.850	1.621	1.794	1.699
	Ebn at Noon		214	221	192	205	244	246	252	238	227	198	213	200
	Edh at Noon		66	71	98	88	56	52	49	59	65	78	62	67
(44) All-Sky Solar Radiation	RadAvg		644	819	923	1190	1391	1417	1495	1480	1386	1102	961	752
	RadStd		153	162	109	141	156	153	101	99	95	94	141	122

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.58	N/A	N/A	+1.28	+1.08	+0.92	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	+1.03	N/A	N/A	N/A	N/A	N/A	+144

Nomenclature: See separate page